

Technology Stack

Date	07 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3	Used to create a responsive, simple, and user-friendly interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Flask provides a lightweight web framework that integrates the trained Machine Learning model and processes prediction requests efficiently.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Files (application_record.csv & credit_record.csv)	The project uses CSV datasets for training and testing the Machine Learning model. No separate database is required for this prototype.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Render	Hosts the Flask web application and provides a public URL for online access.

5	Version Control & CI/CD	Git & GitHub	Git is used for version control, and GitHub is used for source code management, collaboration, and maintaining the project repository.
	(e.g., GitHub, GitLab, Jenkins)		
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost, Jupyter Notebook, VS Code	Scikit-learn is used for model development, Pandas and NumPy for data preprocessing, Matplotlib and Seaborn for visualization, XGBoost for model comparison, and Jupyter Notebook/VS Code for development and implementation.
	(e.g., Stripe, Twilio, Google Maps)		